

§3.2

Measures of Variation

How spread out is the data?

Today's Goal

- Calculate the range, variance, and standard deviation of a dataset
- Distinguish between population and sample formulas for variance and standard deviation
- Apply the Empirical Rule and Chebyshev's Theorem
- Use the coefficient of variation to compare variation across different datasets

What You Will Be Graded On

LT.6 Numerical Summaries of Data

I understand the different numerical summaries for data (e.g., mean, median, mode, standard deviation, etc.). I can find and interpret the values of these numerical summaries for a given data set.

Last time: where the data sits. Today: how far it spreads.

Same Mean, Two Classes

Two statistics classes take the same quiz. Table 3.2.1 in the notes.

Class 1	4	4	4	5	6	6	7	8	8	8	9	9	10	10	10
Class 2	6	6	7	7	7	7	7	7	7	7	8	8	8	8	8

Same mean, 7.2. Which class would you rather teach?



One Number for the Spread

Definition 3.2.4

A **measure of variation** is a number designed to show how much the data values in a data set vary.

The mean could not tell the two classes apart. We will study three measures of variation:

range

variance

**standard
deviation**

Biggest Minus Smallest

Definition 3.2.5

The **range** of a set of quantitative data values is the difference between the maximum data value and the minimum data value:

$$\text{Range} = (\text{max data value}) - (\text{min data value}).$$

Class 1: $10 - 4 = 6$

Class 2: $8 - 6 = 2$

Class 1 has the bigger range, and its dotplot looked more spread out.

Two values in, one number out. Every other score is ignored.

What the Range Buys You

Pros

- Very easy to calculate.
- Provides a quick sense of spread.
- Easy to understand and interpret.

Cons

- Not resistant. Heavily affected by outliers.
- Only uses two data values (min and max).
- Does not reflect variation among all data values.

One wild score, and the range doubles. The other 14 scores never get a vote.

Same Range, Same Spread?

Class 1 again, and a third class.

Class 1	4	4	4	5	6	6	7	8	8	8	9	9	10	10	10
Class 3	4	7	7	7	7	7	7	7	7	7	7	7	7	7	10

Both ranges are 6. Do the two classes have the same spread?

The range looks only at the two ends. Thirteen of Class 3's scores sit at 7. We need a measure that uses every value.

How Far From the Mean, Typically

Definition 3.2.6

The **standard deviation** of a set of quantitative data values is a measure of how much the data values deviate from the mean.

Sample: $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$

Population: $\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$

σ is read “sigma.” Same steps in both. The one difference: a sample divides by $n - 1$, a population by N .

Standard Deviation by Hand

Procedure 3.2.7 Computing the Standard Deviation by Hand

Decide whether the data is a *sample* (use $\bar{x}$ and divide by $n - 1$) or a *population* (use μ and divide by N). The mechanical steps are identical otherwise.

1. Compute the mean ($\bar{x}$ for a sample, μ for a population).
2. For each data value x , compute the deviation $x - \bar{x}$ (or $x - \mu$).
3. Square each deviation to get $(x - \bar{x})^2$.
4. Sum the squared deviations: $\sum (x - \bar{x})^2$.

Standard Deviation by Hand, Continued

Procedure 3.2.7, continued

5. Divide by $n - 1$ (sample) or N (population) to get the variance.
6. Take the square root to get the standard deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}, \quad \sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}.$$

A table with columns x , $x - \bar{x}$, $(x - \bar{x})^2$ keeps the bookkeeping clean for steps 2–4.

Working the Table

Five students from a class of 40 reported their hours studied last week: 3, 5, 6, 8, 8. Find the standard deviation.

Five of 40, so this is a sample. Mean: $\bar{x} = \frac{3+5+6+8+8}{5} = \frac{30}{5} = 6$.

Data x	Deviation $x - \bar{x}$	Squared deviation $(x - \bar{x})^2$
3	$3 - 6 = -3$	$(-3)^2 = 9$
5	$5 - 6 = -1$	$(-1)^2 = 1$
6	$6 - 6 = 0$	$0^2 = 0$
8	$8 - 6 = 2$	$2^2 = 4$
8	$8 - 6 = 2$	$2^2 = 4$
		$\sum (x - \bar{x})^2 = 18$

Finishing the Formula

Sample of $n = 5$, and the squared deviations total 18.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} = \sqrt{\frac{18}{5 - 1}} = \sqrt{4.5} \approx 2.1 \text{ hours}$$

Why divide by 4 when there are 5 values?

We had to estimate the mean from the same five numbers first. That uses up one of them, so a sample divides by $n - 1$. Excel's =STDEV.S does the same.

What the Standard Deviation Buys You

Pros

- Uses every data value in its calculation.
- Same units as the original data.
- Most commonly used measure of variation.
- Larger values indicate greater variation; zero only when all values are the same.

Cons

- Not resistant. Affected by outliers.
- More complex to calculate by hand.
- Sample standard deviation is a (slightly) biased estimator of population standard deviation.
- Value is never negative (cannot indicate direction).

The Square of the Standard Deviation

Definition 3.2.10

The **variance** of a set of quantitative data values is a measure of variation equal to the square of the standard deviation.

Sample: $s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$

Population: $\sigma^2 = \frac{\sum (x - \mu)^2}{N}$

Example 1 again: $s^2 = \frac{18}{4} = 4.5$ square hours. Nobody studies in square hours. That is why we usually report s .

Let Excel Do the Arithmetic

History 101 Quiz 1 scores, from the notes. All 200 scores sit in cells A2 through J21. The 10 sampled scores sit in L2 through L11.

Measure	Sample	Population	Notes
Standard deviation	=STDEV.S (L2 : L11)	=STDEV.P (A2 : J21)	Example 3.2.9
Variance	=VAR.S (L2 : L11)	=VAR.P (A2 : J21)	Example 3.2.11

Where Do Most Scores Sit?

An exam's scores are bell-shaped. The mean is 70 and the standard deviation is 6.

Between which two scores do you expect most of the class?

“Most” needs a number. For bell-shaped data there is a rule that gives one.

The 68, 95, 99.7 Rule

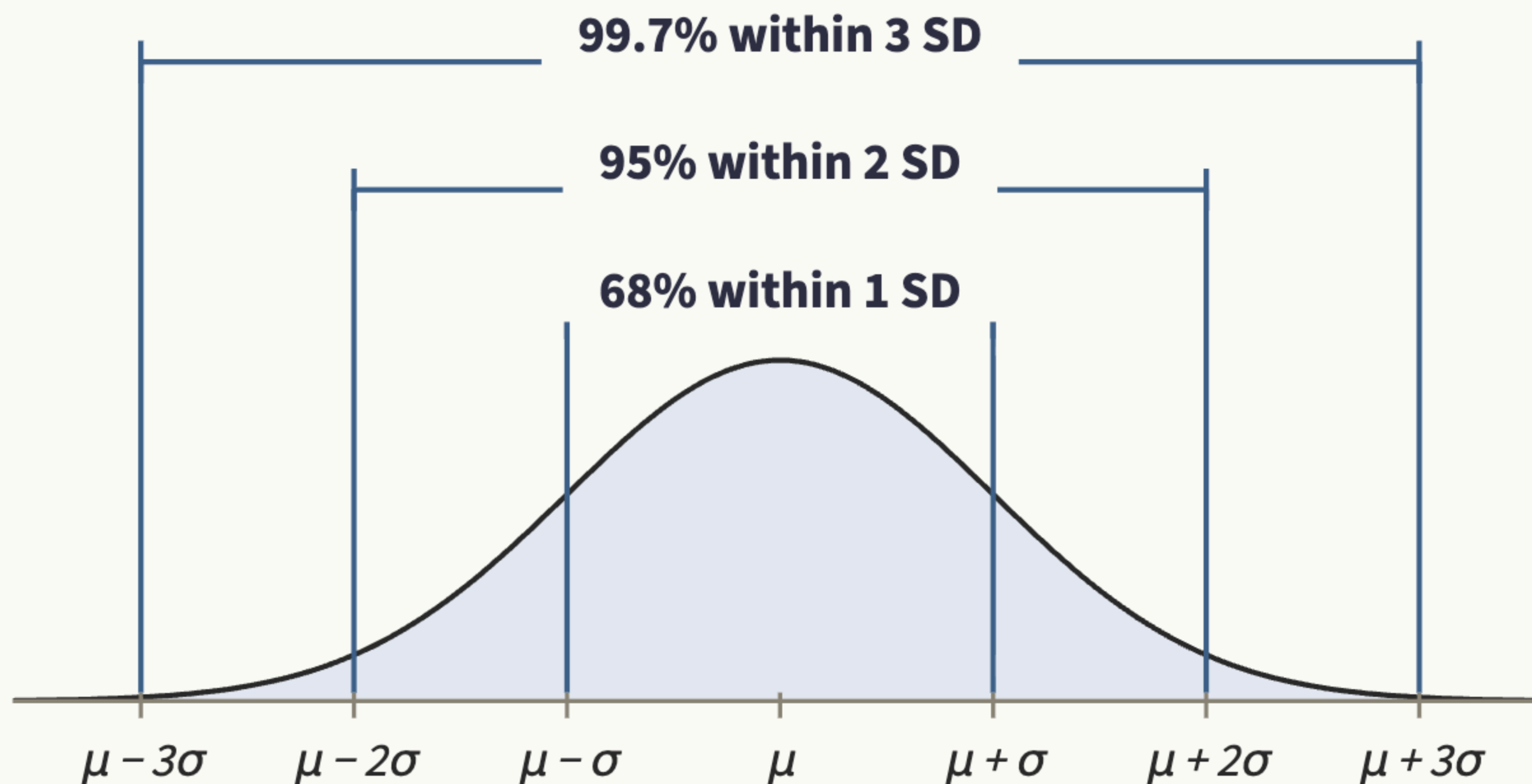
Theorem 3.2.12 The Empirical Rule

If the distribution of a data set is approximately normal (bell-shaped), then

- about 68% of all values fall within 1 standard deviation of the mean,
- about 95% of all values fall within 2 standard deviations of the mean,
and
- about 99.7% of all values fall within 3 standard deviations of the mean.

Mean 70, standard deviation 6: about 95% between 58 and 82. And the first word of the theorem is *if*.

The Rule as a Picture



Redrawn from Figure 3.2.13 in the notes.

When the Data Is Not Bell-Shaped

Not bell-shaped? Then the Empirical Rule does not apply. Chebyshev's Theorem still does.

Theorem 3.2.14 Chebyshev's Theorem

The proportion of any data set lying within k standard deviations of the mean is always *at least* $1 - \frac{1}{k^2}$, where k is any positive number greater than 1.

Any shape at all. The price: it promises only *at least*. The true percentage is usually higher.

Chebyshev Versus the Empirical Rule

A data set has mean 70 and standard deviation 6.

(a) According to Chebyshev's Theorem, at least what percentage of the values must lie between 58 and 82?

$58 = 70 - 2(6)$ and $82 = 70 + 2(6)$, so $k = 2$. $1 - \frac{1}{2^2} = \frac{3}{4} = 75\%$. **At least 75% of the values must lie between 58 and 82.**

(b) Suppose the data set is bell-shaped. According to the Empirical Rule, about what percentage of the values lie between 58 and 82?

58 and 82 are 2 standard deviations from the mean. About 95% of the values lie between 58 and 82.

Chebyshev Versus the Empirical Rule

Same data set: mean 70 and standard deviation 6.

(c) According to Chebyshev's Theorem, at least what percentage of the values must lie between 52 and 88?

$$52 = 70 - 3(6) \text{ and } 88 = 70 + 3(6), \text{ so } k = 3. 1 - \frac{1}{3^2} = \frac{8}{9} \approx 88.9\%.$$

At least 88.9% of the values must lie between 52 and 88.

(d) Suppose the data set is bell-shaped. According to the Empirical Rule, about what percentage of the values lie between 52 and 88?

52 and 88 are 3 standard deviations from the mean. About 99.7% of the values lie between 52 and 88.

Back to the Two Classes

Each class is a whole population, so we compute σ . Table 3.2.16 in the notes.

	Class 1	Class 2
μ	7.2	7.2
σ	2.2	0.7

Since σ is higher for Class 1 than it is for Class 2, the quiz scores in Class 1 are more spread out than in Class 2.

Compare standard deviations only when scale, units, and (roughly) means match. Otherwise, read on.

Both Have a Standard Deviation of 1

Quiz Score	Exam Score
7	77
8	78
9	79

Both standard deviations are 1. Same variation?

One point is 10% of a 10-point quiz but 1% of a 100-point exam. The quiz scores have higher variation than the exam scores.

Spread as a Percent of the Mean

Definition 3.2.18

The **coefficient of variation** of a data set is the ratio of the standard deviation to the mean:

$$CV = \frac{s}{\bar{x}} \quad (\text{for a sample})$$

$$CV = \frac{\sigma}{\mu} \quad (\text{for a population}).$$

The coefficient of variation is normally reported as a percentage, rounded to one decimal place.

Computing the Coefficient of Variation

Procedure 3.2.19 Computing the Coefficient of Variation

For each data set:

1. Compute the mean ($\bar{x}$ for a sample, μ for a population).
2. Compute the standard deviation (s for a sample, σ for a population).
3. Compute the ratio $\frac{s}{\bar{x}}$ (or $\frac{\sigma}{\mu}$).
4. Multiply by 100% and round to one decimal place.

Then compare the CVs across data sets. The larger CV indicates greater relative variation.

Sales or Customers: Which Varies More?

A coffee shop recorded two things on each of 30 days: its total sales for the day, and the number of customers it served that day.

Daily sales: $\bar{x} = \$900$, $s = \$90$. Customers per day: $\bar{x} = 200$, $s = 30$.

(a) Find the coefficient of variation for daily sales.

$$CV = \frac{\$90}{\$900} \cdot 100\% = 10.0\%$$

(b) Find the coefficient of variation for customers per day.

$$CV = \frac{30}{200} \cdot 100\% = 15.0\%$$

Sales or Customers: Which Varies More?

Daily sales: $CV = 10.0\%$. Customers per day: $CV = 15.0\%$.

(c) Which has more variation, daily sales or customers per day?

Customers per day. Its CV is 15.0%, and the CV for daily sales is 10.0%.

The standard deviations cannot be compared directly. One is in dollars, the other in people.

Four Measures of Spread

Measure	Reach for it when	Watch out
Range	You want a quick sense of spread.	Uses only two values.
Standard deviation	Almost always. Same units as the data.	Sample or population? Pick the formula.
Variance	A later formula asks for it.	Squared units.
Coefficient of variation	Different units or scales.	Needs a mean well away from zero.

Rule of thumb. Report the standard deviation with the mean. Reach for the CV only when the two data sets are not on the same footing.

A Standard Deviation of Zero

**Twenty values,
standard deviation 0.
What do you know?**

Every deviation is 0, so all 20 values equal the mean. Same number, twenty times.

Could s ever be negative?

No. It is the square root of a sum of squares. Zero is the floor.